ULTRAFILTRATION

Group #3
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Chemical Engineering Lab II

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Introduction

- The purpose of this lab is to examine the effectiveness of an ultrafiltration system.
  - *Used by determining permeate flux and different inlet and outlet pressures, and at different recoveries, and determining the optimum pressure settings for optimal flux.*
- Used in industrial applications to purify a solvent by removing solute from a solution, or vis versa. As well as separating two solutes by using membrane permeable to only one of the solutes.
  - *In this experiment, a milk-water mixture is filtered to concentrate the milk (solute).*

$h_L = K \frac{}{2}$

Theory

- Underlying equations:
  - *Concentration Factor :* $CF = \frac{V_{orginal}}{V_{orginal} - V_{permeate}}$
  - *Percent Recovery:* $R = \frac{V_{permeate}}{V_{orginal}} * 100\%$
  - *Membrane Permeability* $Q_v = \frac{\nu}{\Delta P}$ *Where* ν *= is volume flux*

Apparatus

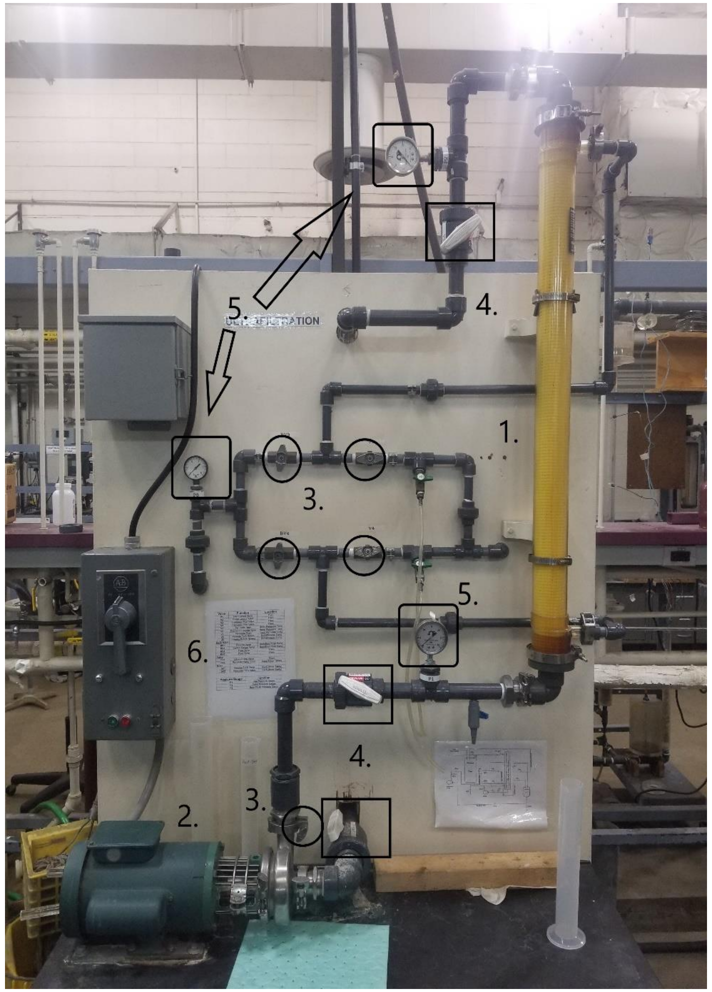

Figure 1: Ultrafiltration Apparatus

Table 1: Description of Key Components and how they are Used in Figure 1

	Component	Description
1.	Hollow Fiber Membrane	Separates the milk proteins from the solution.
1.	Circulation Pump	Pumps the solution to the system.
1.	Flow Control Valves	Control the flows (all circled).
4.	Pressure Valves	White valves that control pressure (boxed).
5.	Pressure Gauges	Reads the pressure in psi (rounded boxes).
6.	Power Supply	Supplies power to the system.

Apparatus continued

Figure 2: Process Tank

Figure 3: Permeate Tank

Table 2: Description of Key Components and how they are Used in Figure 2

	Component	Description of Component
1.	Process Tank	The milk solution is stored in the tank before being pumped into the system.
2.	Back-Flush Pump	Used in the cleaning of the system
3.	Flow Control Valves	Control the Flow (Valves circled)

Table 3: Description of Key Components and how they are Used in Figure 3

	Component	Description of Component
1	Permeate Tank	The various solutions are stored in the tank before being pumped into the system.
2	Flow Control Valves	Control the flow (valves circled).

Apparatus continued

Figure 4: Drainage system

Table 4: Description of Key Components and how they are Used in Figure 4

	Component	Description of Component
1.	Drainage valve for the system	Drains the filtration part of the system.
2.	Drainage valve for the process and permeate tanks.	Drains the tanks upon completion.

Figure 5: Temperature Analog and DI Water Flow controls

Table 5: Description of Key Components and how they are Used in Figure 2

	Component	Description of Component
1.	Analog Meter	Reads the temperature of the DI water coming into the system.
2.	DI Needle Valve	Controls the flow of the DI water to the steam heat exchanger.
3.	Flow Control Valve	Controls the flow of the heated DI water to the process tank.

Materials and Supplies

Table 6: Materials and Supplies

	Material/Supply	Description
1.	Milk	Will be mixed with DI water
2.	Deionized water	Used to clean and make the solution
3.	Phosphoric Acid	Used to clean the system
4.	Sodium Hydroxide	Used to clean the system
5.	pH probe	Used to measure the pH of acid solution
6.	Scale	Used to measure chemicals for the various solutions
7.	Beaker	Used in the making of the solutions
8.	Stopwatch	Used to measure time intervals
9.	Goggles	Used for safety
10.	Gloves	Used for safety
11..	Mop and Bucket	Used to for spills
12.	Graduated Cylinder	Used to measure collected permeate

Clean and Preparation Procedure

- Cleaning (Tuesday section): During pre-lab week
  - *1. Liquid Solution Flush: Circulate 20L of DI water at 105 F for 10-15 mins at given inlet and outlet pressures.*
  - *2. Acid Cycle : 20L of pH 2.00 Phosphoric acid with DI Water at 105 F for 10 minutes (same conditions as Liquid Solution Flush).*
  - *3. Drain and Rinse Cycle: Circulate 50L of DI Water at 105F for 10-15 minutes (same conditions as Liquid Solution Flush).*
- Preparation: Lab week
  - *1. Sanitation Cycle: Circulate 200 ppm sodium hypochlorite solution in 20 L of DI Water at 105 F 10 -15 minutes (same conditions as Liquid Solution Flush).*
- Clean up (Post Lab)
  - *Repeat Drain and Rinse Cycle (3) before leaving lab.*

Lab Procedure

- 1. Water Flux Test : Use 20L of DI water at 105 F.
  - *Determine permeate water flux at an inlet pressure of 25 psi and outlet pressures of 5, 10, 20 and 15 psi.*
- Milk Test
  - *1. Fill tank with 80 L of 105 F DI water and add 1 gallon of milk and stir.*
  - *2. Start system and adjust inlet pressure to 25 psi and 5 psi outlet.*
  - *3. Run until 10% recovery is reached (10% original volume removed into permeate tank).*
  - *4. Return the permeate line to the process tank (no recovery).*
  - *5. Measure permeate flux (volume of permeate flow in certain amount of time ~10s) with different outlet pressures (5, 15, 10, 20, and 5 psi)*
  - *6. Repeat steps 3-5 at recoveries of 25% and 40%.*

Experimental Challenges

- Reading the volume of the process tank when altering recovery percentages.
  - *Use a ruler to approximate volume of milk solution removed.*
- Consistent flow of permeate when measuring permeate flux.
  - *Let permeate flow for a few seconds before starting the measurement.*
- Inlet and outlet pressure altering when taking measurements.
  - *Have someone be responsible for watching pressures and adjust accordingly.*
- Keeping filter tube filled when altering pressures.
  - *Close valve 3 occasionally to allow the filter to be completely filled.*

Safety

- Do not overfill the tank
- Do not run the pump with no water/ solution in the tank.
- Be cautious of water on the floor from small leaks throughout the system, can lead to slippery floor conditions.
- Do not adjust inlet and outlet pressure valves too quickly.
- Wear proper chemical safety gloves when dealing with the concentrated acid and bases.